

From Numbers to Narrative: Augmenting SHAP Explanations with Large Language Models for Robotic Choreography Feedback

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1 Introduction

Building upon the predictive models and SHAP-based interpretability framework established in our initial study, this project extension addresses a crucial challenge in Human-Robot Interaction (HRI): translating complex machine learning explanations into actionable, human-centric insights. While SHAP values provide a precise, quantitative measure of each feature’s impact on a model’s prediction, their numerical nature can be opaque to domain experts such as choreographers and designers who lack a background in data science.

The primary goal of this extension is to bridge this "interpretability gap." We achieve this by integrating a Large Language Model (LLM) into our analysis pipeline. This new module leverages the local SHAP explanations and raw feature data to automatically generate natural language feedback. The key objectives were:

1. **Generate Explanations:** Automatically produce clear, textual feedback explaining why a specific choreography received its predicted evaluation score.
2. **Summarize Key Drivers:** Create human-readable summaries of the most influential features identified by SHAP.
3. **Provide Actionable Suggestions:** Offer tailored recommendations for choreographers to enhance specific aspects of their robotic performances.

By transforming numerical data into narrative feedback, this work makes our predictive models more accessible and useful, fostering a more intuitive and data-informed design cycle for robotic choreography.

2 System Architecture and Methodology

To achieve our objectives, we designed and implemented a pipeline that takes a single choreography instance as input and produces a multi-faceted textual analysis as output. The system is delivered through an interactive Gradio-based web interface, allowing users to explore random instances and generate feedback on demand.

2.1 System Pipeline

The core of the system is a three-stage pipeline:

1. **Data Ingestion:** For a given choreography, the system gathers three key pieces of information: the raw feature values (e.g., `nMovements=15`, `timeDuration=120s`), the model’s predicted score, and the corresponding local SHAP values for each feature.
2. **Prompt Engineering:** This data is then structured into a sophisticated prompt for the LLM. The prompt is designed to provide comprehensive context and constrain the output format, ensuring relevance and clarity.
3. **LLM-Powered Generation:** The structured prompt is sent to an LLM (Google’s Gemini Pro model), which generates the final textual analysis. This output is then displayed to the user.

2.2 Prompt Engineering: The Core Component

The quality of the LLM's output is critically dependent on the quality of the prompt. Our final prompt design incorporates several key strategies to ensure high-fidelity explanations:

- **Role-Playing:** The LLM is instructed to act as an expert in both machine learning explainability and HRI, setting the correct tone and domain context.
- **Global Context Injection:** The prompt is pre-loaded with high-level insights from our initial SHAP analysis. It explicitly states that `nMovements`, `timeDuration`, and `musicBPM` are the most dominant features and that their effects are often interactive and contextual. This grounds the local explanation within the global model behavior.
- **Few-Shot Learning:** The prompt includes two complete, high-quality examples of analysis (one for a high-scoring and one for a low-scoring instance). This "few-shot" approach teaches the LLM the desired style, structure, and depth of analysis.
- **Structured Output Mandate:** The LLM is required to generate its response in a fixed five-part format: 1) *Key Insights*, 2) *Feature Analysis*, 3) *Model Behavior*, 4) *Practical Implications*, and 5) *Surprising Findings*. This ensures that every explanation is comprehensive and consistent.

A condensed version of the prompt template provided to the LLM is shown below:

```
1 [SYSTEM]
2 You are an expert in ML explainability and HRI. Your task is to analyze a prediction for
  a robot choreography.
3 General Context: 'nMovements', 'timeDuration', and 'musicBPM' are the key drivers.
  Feature interactions are crucial. A high feature value is not always "good"; its
  effect is contextual.
4 Required Output Structure:
5 1. Key Insights: One-sentence summary.
6 2. Feature Analysis: Top 3-5 influential features.
7 3. Model Behavior: Generalize what the model learned.
8 4. Practical Implications: Actionable advice.
9 5. Surprising Findings: Any unexpected contributions.
10 ---
11 [EXAMPLE 1: High Score Analysis...]
12 ---
13 [EXAMPLE 2: Low Score Analysis...]
14 ---
15 [YOUR TURN: Analyze the Following Instance]
16 - Model Prediction: {prediction_value}
17 - Features:
18   - nMovements: {value}
19   - timeDuration: {value}
20   ...
21 - SHAP Values:
22   - nMovements: {shap_value}
23   - timeDuration: {shap_value}
24   ...
```

Listing 1: Condensed LLM Prompt Structure

2.3 Implementation

The system was implemented in Python using the `gradio` library for the user interface and Google's `generativeai` library to interface with the Gemini API. A custom class, `SHAPInstanceExplorer`, manages data loading, API configuration, prompt creation, and interaction logic. The interface allows users to select a model type (classification/regression) and a target metric, load a random choreography instance, view its features and SHAP values, and click a button to generate the LLM-based analysis.

3 Results: Generated Feedback Example

The implemented system successfully generates detailed, relevant, and actionable feedback. The structured prompt forces the LLM to provide nuanced explanations that account for feature interactions,

moving beyond simplistic interpretations. Figure 1 showcases a detailed example of the generated feedback for a specific choreography instance that received a low score for the **Human Characterization** metric.

Figure 1: An example of LLM-generated feedback for a low-scoring instance on ‘HumanCharacterization’. The system analyzes the key feature and SHAP values to produce actionable insights and a detailed narrative explanation.

Analysis of a Low-Scoring Choreography (Instance ID: 632)	
Input Data	Value
Predicted Score (Classification)	0.0 (Low Characterization)
timeDuration (Feature Value / SHAP Value)	144s / -0.200
nMovements (Feature Value / SHAP Value)	20 / -0.078
musicBPM (Feature Value / SHAP Value)	94 / +0.048
movementsTransitionsDuration (Feature / SHAP)	2 / -0.020

LLM-Generated Analysis 1. Key Insights: The model predicted a low ‘Human Characterization’ score primarily because the relatively long duration and high number of movements, when combined with a slower music BPM, may have created a performance that felt overwhelming and less relatable.

2. Feature Analysis:

- **timeDuration** (Value: 144s, SHAP: -0.200): This was the strongest negative contributor. While duration is generally beneficial, in this case, the extended length detracted from the human characterization. This could be due to the content not being engaging enough to hold attention for that long.
- **nMovements** (Value: 20, SHAP: -0.078): The high number of movements also negatively impacted the score. The model might perceive this as being too complex or disjointed, hindering the development of a clear and relatable persona.
- **musicBPM** (Value: 94, SHAP: +0.048): The music BPM contributed positively, partially offsetting the negative impacts of other features. The slightly slower BPM may have allowed the individual movements to be more clearly defined and expressive.
- **movementsTransitionsDuration** (Value: 2, SHAP: -0.020): A medium transition duration further detracted from the score, indicating that the pace or flow between movements might not have been optimized for human characterization.

3. Model Behavior: The model appears to penalize choreographies that, while lengthy and complex, lack a cohesive or engaging human element. It may be sensitive to the relationship between duration, complexity, and the ability to create a relatable performance.

4. Practical Implications: To improve a robot’s ‘Human Characterization’ score, a choreographer should:

- *Optimize Duration:* If the choreography is lengthy, ensure there is enough engaging content to justify the duration. Consider shortening the routine or adding more captivating elements.
- *Simplify Movements:* Reduce the number of distinct movements to create a more focused and understandable performance.
- *Improve Transitions:* Focus on creating smoother and more meaningful transitions between movements.

5. Surprising Findings: It is surprising that ‘movementsDifficulty’ (level 3) and ‘humanMovements’ (level 2) had almost no impact on the prediction. This implies that for this specific choreography, the model placed far more importance on the overall structure and pacing of the performance than on the technical difficulty or the presence of human-like movements. This suggests that audience characterization is more dependent on relatability and engagement than technical skill.

4 Discussion and Future Work

This extension successfully demonstrates the potential of LLMs to serve as a “translation layer” between quantitative model explanations and qualitative user insights. The structured, context-aware prompt engineering proved to be the most critical factor in achieving high-quality, relevant, and non-generic feedback.

However, some challenges remain. A key challenge in operationalizing this pipeline is managing computational resources, particularly the token count sent to the LLM. To mitigate this, we can implement a

strategy of truncating less meaningful numerical data, such as SHAP values with high precision, thereby reducing the input token size without sacrificing critical information. Furthermore, while our prompting strategy minimizes the risk, LLMs can occasionally "hallucinate" or generate plausible but incorrect explanations. This risk is amplified because our current implementation provides only local (instance-specific) SHAP values. Without explicit data on global feature importance or interaction effects, the LLM might incorrectly infer or invent the reasons behind complex feature relationships.

Future work could advance this project in several exciting directions:

- **Real-Time Interactive Tuning Interface:** Develop a more dynamic UI where users can adjust input features—such as `nMovements` or `musicBPM`—using sliders or input boxes. This would allow for real-time prediction updates, providing an immediate, intuitive understanding of how different choreographic choices impact evaluation scores.
- **Integration of Multimodal LLMs for Visual Explanation:** Leverage the capabilities of multimodal LLMs (like GPT-4V or Gemini) to directly interpret plots. This would enable feeding global explanation graphics, such as SHAP summary or dependence plots, into the prompt, asking the model to synthesize insights from both visual and numerical data for a more holistic explanation.
- **Development of a Public Web Framework:** Transition the prototype from a demonstration tool into a polished web application with standardized metrics and features. By making it publicly available, this project could become a valuable framework for the HRI and creative robotics community to benchmark and interpret their own models.

5 Conclusion

This project extension successfully designed and implemented a novel pipeline that integrates Large Language Models with SHAP-based explainability. By translating complex numerical SHAP values into structured, narrative feedback, our system makes the reasoning of our predictive models accessible to non-technical stakeholders. The resulting interactive tool provides choreographers with not only a prediction of performance quality but also a clear, actionable explanation of the "why" behind it. This work represents a significant step towards creating more collaborative and intuitive human-AI systems for creative domains.